Some of contents:

- Smooth Manifold;
- De Rham Cohomology;
- Integratiability, Foliation;
- Differential Geometry without Metric(Connection Geometry);
-

And the following contents are not contained:

- Specified Geometry;
- Characteristic Classes;
- Algebraic Topology: Manifolds;
- Chern-Weil;
- Analysis on Manifolds;

Chapter 1 Smooth Manifold

1.1 Definitions

Definition 1.1.0.1 An n-dimensional **smooth manifold** is a topological manifold $(M, \{(U_i, \varphi_i)\})$ with **smooth transition function**

$$\varphi_i \circ \varphi_j^{-1} \in C^{\infty}_{diff}(\hat{U}_j, \hat{U}_i)$$

And for every smooth charts, there exists a **maximal chart** which contains the smooth charts. Such atlas is called the **smooth structure** of M.

◇ **Lemma 1.1.0.2** Let M be a topological manifold:

- Every smooth atlas is contained in a maximal smooth atlas;
- Two smooth atlases for M determines the same maximal smooth atlas iff their union is smooth.

∘ **Proof**

- Let $\mathcal{A}$ be the collection of charts that are smooth compatible with every chart in $\bar{\mathcal{A}}$.

  $\bar{\mathcal{A}}$ is a smooth chart. Firstly, $\mathcal{A} \subseteq \bar{\mathcal{A}}$. For two charts $(U, \varphi), (V, \psi) \in \bar{\mathcal{A}}$ where $U \cap V \neq \varnothing$, select some $p \in U \cap V$ and a chart (W, θ) where $p \in U \cap V \cap W$. Omitted.
- Omitted.

□

▶ ***Theorem*** (Milnor,1956) S^7 has more than one the smooth structures.

▶ ***Theorem*** (Exotic $\mathbb{R}^4$) $\mathbb{R}^4$ has infinite many of smooth structures.

1.1.1 Smooth Functions

Definition 1.1.1.1 A function $f : M \to \mathbb{R}$ is **smooth function** if for every coordinate (U_i, φ_i)

$$f \circ \varphi_i^{-1} : \hat{U}_i \to \mathbb{R}$$

is smooth.

1.1.2 Construction of Smooth Manifold

◇ **Lemma 1.1.2.1** (Smooth Manifold Construction Lemma) Let M be a set, and suppose we are given a collection $\{U_\alpha\}$ of subsete of M, together with injective maps $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$. With following properties:

- For each α, $\varphi_\alpha(U_\alpha)$ is an open subset of $\mathbb{R}^n$;
- $\varphi_\alpha(U_\alpha \cap U_\beta)$ are open in $\mathbb{R}^n$;
- Whenever $U_\alpha \cap U_\beta \neq \varnothing$, $\varphi_\alpha \circ \varphi_\beta^{-1}$ is a diffeomorphism;
- Countably many U_α cover M;

- Hausdorfness;

∘ **Proof** Omitted. □

1.2 Smooth Functions and Smooth Maps

1.2.1 Definitions

Definition 1.2.1.1

- A function $f : M \to \mathbb{R}^k$ is **smooth** if for every $p \in M$, there exists a smooth chart (U, φ) and the composition $f \circ \varphi^{-1} : \hat{U} \to \mathbb{R}^k$ is smooth;

  Such $f \circ \varphi^{-1}$ is called the **coordinate representation** of f;

- Furthermore, $F : M \to N$ is smooth if for some charts $(U, \varphi), (V, \psi)$,

$$\varphi \circ F \circ \psi^{-1} : \psi(U \cap V) \to \varphi(U \cap V)$$

  is smooth

It is easy to verify:

- $C^\infty(M)$ is commutative $\mathbb{R}$-algebra;
- For every charts, the representation map is smooth;

◇ **Lemma 1.2.1.2** (Smoothness is Local)Let M, N be two smooth manifolds, $F : M \to N$ be a map. If every point $p \in M$, restriction $F|U$ is smooth, then F is smooth; Conversely also holds.

∘ **Proof** Omitted; □

◇ **Lemma 1.2.1.3** (Pasting) If given $F_\alpha : U_\alpha \to N$ with $F_\alpha|U_\alpha \cap U_\beta = F_\beta|U_\alpha \cap U_\beta, \cup U_\alpha = M$. Then there exists $F : M \to N$ such that

$$F|U_\alpha = F_\alpha$$

∘ **Proof** ... □

◇ **Lemma 1.2.1.4** Every smooth map between smooth manifolds is continuous.

∘ **Proof** Omitted. □

1.2.2 Diffeomorphisms

Definition 1.2.2.1 ...

1.3 Smooth Covering Maps

1.4 Proper Maps

Definition 1.4.0.1 $F : M \to N$ is **proper** if the inverse image of any compact $K \subseteq N$ is compact.

◇ **Lemma 1.4.0.2** Suppose M is a compact space and N is a Hausdorff space, then every continuous map $F : M \to N$ is proper.

∘ **Proof** ... □

1.5 Partition Unity

Chapter 2 Tangent Spaces

2.1 Tangent Vectors

2.1.1 Tangent Space as Derivation of the Space of Germs

Germs

Definition 2.1.1.1

- The **germ** on $C_p^\infty(M)$ is specified by the equivalence class by:

$$(U, f) \simeq (V, g) \Leftrightarrow \exists W, p \in W \subseteq U \cap V, f|W = g|W$$

  And operations

  ...

  Thus is a linear space;

- A **derivation** of $C_p^\infty(M)$ is a linear map

$$v : C_p^\infty(M) \to \mathbb{R}$$

  which satisfies Lebnitz's law:

  ...

2.1.2 Tangent Vectors as Equivalence Classes of Vectors

Definition 2.1.2.1 ...

2.2 Differentials

2.2.1 Derivative on Opens

Definition 2.2.1.1 ...

2.2.2 Formel Derivations

Definition 2.2.2.1 Let M be a manifold (with or without boundary), then a linear map

$$C^\infty(M)^* \ni v : C^\infty(M) \to \mathbb{R}$$

such that:

- (Lebnitz's Law) $v(fg) = f(p)v(g) + g(p)v(f)$;

♠ **Proposition 2.2.2.2** The follwoings vector spaces are isomorphic:

-
-

•

∘ **Proof** ... □

2.2.3 Differential of Smooth Map

Definition 2.2.3.1 ...

2.2.4 Pushforwards

2.3 Coordinate Computations

2.4 Tangent Vectors to Curves,Fibres

2.4.1 Curves

2.4.2 Fibres

2.5 Tangent Bundles

Definition 2.5.0.1 Define: For manifold M, the **tangent space**

$$TM = \coprod_{p \in M} T_p M$$

Then:

♠ **Proposition 2.5.0.2** (Smooth Structure of TM, Smoothness of Bundle Map)

∘ **Proof** ... □

2.5.1 Functoriality

♠ **Proposition 2.5.1.1** $T : Man^1 \to VBun$ is a functor.

∘ **Proof**

- Firstsly, T is well-defined;
- Secondly, $T(g \circ f) = T(g) \circ T(f)$. $f : M \to N, g : N \to L$,

$$\begin{aligned} T_p(g \circ f)(X)(h) &= X(h \circ (g \circ f)) \\ &= T_p(f)(X)(h \circ g) \\ &= T_p(g) \circ T_p(f)(X) \end{aligned}$$

where $X \in T_p M$.

□

Especially

♠ **Proposition 2.5.1.2** If $f : M \to N$ is smooth map, so is $T(f)$.

∘ **Proof** Consider followin diagram

$$\begin{array}{ccc} TM & \xrightarrow{T(f)} & TN \\ \downarrow & & \downarrow \\ M & \xrightarrow{f} & N \end{array}$$

Take charts □

2.5.2 Pfaffian Form

Definition 2.5.2.1 Let M be a differentiable manifold and V a finite dimensional vector space. A **V-valued Pfaffian form**, also called 1-form on M is a smooth map

$$\omega : TM \to V$$

where $\omega_p : T_pM \to V$ is linear.

The set of all Pfaffian forms on M

$$\Omega^1(M, V)$$

Especially, when $V = \mathbb{R}$, denoted by $\Omega_1(M)$. As a matter of fact

$$\Omega^1(M, V) = \{\omega | \omega_p : T_pM \to V, p \in M\}$$

Definition 2.5.2.2 Define the product

$$C^\infty(M) \times \Omega^1(M, V) \to \Omega^1(M, V)$$
$$f \cdot \omega := (f \circ \pi_{TM}) \cdot \omega$$

♣ **Example 2.5.2.3** ...

Pullback

Definition 2.5.2.4 Let $f : N \to M$. Then, the **pullback**

$$f^* : \Omega^1(M, V) \to \Omega^1(N, V)$$
$$\omega \mapsto \omega \circ (Tf)$$

Pfaffian Form with Vector Bundles

Consider the vector bundle:

$$\pi : E \to M \qquad \tau : T^*M \to M$$

Notice that:

$$\text{(Locally Trivial)} \qquad \pi^{-1}(U) \cong U \times \mathbb{R}^n$$

Then, define:

$$\Omega^1(M;E) = \Gamma(T^*M \otimes E)$$

And we observe the section locally, with local triviality $\pi^{-1}(U) \cong U \times \mathbb{R}^n$, we have

$$\Gamma(T^* \otimes E)|U = \Omega(U;\mathbb{R}^n)$$

2.5.3 Curve, and its Derivative

Definition 2.5.3.1

- A **smooth curve** is a smooth function

$$\gamma : I \to M$$

- If γ could be seen as a smooth map between manifolds I and M, now under the standard basis of I, we obtain

$$(d\gamma)_t : T_t I = \mathbb{R} \to T_{\gamma(t)} M$$

  Especially, with the basis $(\frac{\partial}{\partial t}|t_0)$, define

$$\dot{\gamma} : I \to TM \qquad t \mapsto (d\gamma)_{t_0}(\frac{\partial}{\partial t}|t_0) \in T_{\gamma(t_0)}M$$

♠ **Proposition 2.5.3.2**

- We have:

$$\tau \circ \dot{\gamma} = \gamma$$

- And $\dot{\gamma}$ is actually the section of pullback bundle: The pullback bundle induced by γ:

$$\begin{array}{ccc} \gamma^*TM & \dashrightarrow & TM \\ \vdots & & \downarrow \\ I & \longrightarrow & M \end{array}$$

  Then, $\dot{\gamma} \in \gamma^*TM$.

∘ **Proof**

- Definition;
- Omitted;

□

2.6 Vector Fields and Tensor Fields

Definition 2.6.0.1 Let M be a manifold.

- A **vector field** on M a section of the map $\pi : TM \to M$.
- More concretely, is a continuous map $X : M \to TM$, with

$$\pi \circ X = Id_M$$

Denoted by:

- $\mathfrak{X}(M)$; or
- $\Gamma(M, TM)$; or
- $H^0(M, \mathcal{T}_M) = \Gamma(M, \mathcal{T}_M)$; For sheaf $\mathcal{T}_M(U) := \Gamma(U, TU)$;

Equivalently, this is a continuous map with $X_p \in T_pM$;

♠ **Proposition 2.6.0.2** ...

∘ **Proof** ... □

Smoothness of Vector Field

♠ **Proposition 2.6.0.3** (Smooth Vector Fields as $C^\infty(M)$-Module)

∘ **Proof** ... □

Extension of Vector Field

Local and Global Frames

2.6.1 Vector Fields as Derivation of $C^\infty(M)$

Definition 2.6.1.1 Define the action of $X \in \mathfrak{X}(M)$:

$$X : C^\infty(M) \to C^\infty(M) \qquad f \mapsto (Xf : p \mapsto (Xf)(p) = X_p f)$$

♠ **Proposition 2.6.1.2** For $X \in \mathfrak{X}(M)$; $f, g \in C^\infty(M)$ we have

$$X(fg) = X(f)g + X(g)f \qquad X_p(fg) = X_p(g)f(p) + X_p(f)g(p)$$

∘ **Proof** ... □

♠ **Proposition 2.6.1.3** Conversely, given $D \in Der_{\mathbb{R}}(C^\infty(M), C^\infty(M))$, then

$$X : M \to TM \qquad X_p(f) := (Df)(p)$$

∘ **Proof** We should verify: X_p is the element of T_pM. □

★ **Corollary 2.6.1.4** There exists an isomrophsim

$$\mathfrak{X}(M) \cong Der_{\mathbb{R}}(C^\infty(M), C^\infty(M))$$

∘ **Proof** Omitted. □

2.6.2 Lie Derivatives

◇ **Lemma 2.6.2.1** If $U \subseteq M$ is an open subset and $f \in C^\infty(U, V)$ is a smooth function on U with values in V, where V is a finite dimensional vector space, $X \in \mathcal{V}(M)$.

Then we obtain for each a smooth function

$$\mathcal{L}_X f := df \circ X | U \in C^\infty(U, V)$$

∘ **Proof** Omitted. □

Definition 2.6.2.2 Such $\mathcal{L}_X(f)$ is called **Lie derivative** of f with respect to X.

2.6.3 Lie Bracket

Definition 2.6.3.1 Define the **Lie bracket** on $\mathfrak{X}(M)$: For $X, Y \in \mathfrak{X}(M)$,

$$[X, Y] : C^1(M) \to C^\infty(M) \qquad [X, Y]f := X(Yf) - Y(Xf)$$

♠ **Proposition 2.6.3.2** $(\mathfrak{X}(M), [-,-])$ is a Lie algebra. And $[X, Y]$ is smooth.

∘ **Proof** ... □

♠ **Proposition 2.6.3.3** (Naturality)

∘ **Proof** ... □

Coordinate Calculating of Lie Bracket

2.7 Tangent Covectors on Manifolds

Definition 2.7.0.1 Let M be a manifold. For each $p \in M$, we define the **cotangent space** at p,

$$T_p^* M := (T_p M)^\vee$$

2.7.1 Conservative Covector Fields

Chapter 3 Bundles

Some of contents should refer to the note: Bundles. We will discuss the bundles in general there. Others: Principle Bundles, Associated Bundles, usw.

3.1 Vector Bundles

Definition 3.1.0.1 ...

$C^\infty(U)$-Module Structure

Definition 3.1.0.2 The vector bundle $s \in \Gamma(U)$ has a $C^\infty(U)$-structure: For $f \in C^\infty(U)$,

$$f \cdot s \in C^\infty(U)$$

3.1.1 Vector-Bundle Valued Differential Forms

Definition 3.1.1.1 Let $\pi : E \to M$ be a vector bundle. Now, there exists two bundles

$$\pi : E \to M \qquad \tau : TM \to M \Rightarrow \Lambda^k \tau \otimes \pi$$

Define:

$$\Omega^k(M; E) := \Gamma(\Lambda^k T^* M \otimes E)$$

Especially: We will not define the differential here, to make $(\Omega^i(M; E))$ a chain complex.
Because: There exists no differential d without a structure of connection. We will discuss thsi later.

$\Omega^0(M; E)$

We have $\Omega^0(M, E) = \Gamma(E)$;

$\Omega^1(M; E)$**,** $\Omega^k(M; E)$

♠ **Proposition 3.1.1.2** For vector bundle $\pi : E \to M$:

$$\Omega^1(M; E) = \Gamma(T^* M \otimes E) \cong Hom_{C^\infty(M)}(\mathfrak{X}(M), \Gamma(E))$$
$$\Omega^k(M; E) = \Gamma(\Lambda^k T^* M \otimes E) \cong Alt^k_{C^\infty(M)}(\mathfrak{X}(M), \Gamma(E))$$

∘ **Proof** We just need to verify, that

$$\Gamma(\Lambda^k T^* M \otimes E) \cong Alt^k_{C^\infty(M)}(\mathfrak{X}(M), \Gamma(E))$$

Because in linear algebra:

$$\Lambda^k V^* \otimes W \cong Alt^k(V, M)$$

□

Local Frame Calculating

3.1.2 Fibre, and Operations by Fibrewise Construction

3.1.3 Bundle Metric

Definition 3.1.3.1 Let $\pi : E \to M$ be a vector bundle, then, the **bundle metric** is

$$k : E \times_M E \to M \times \mathbb{R}$$

where the restriction of k on the fibre of M is a bilinear map over vector space. This implies:

$$k|\pi^{-1}(m) \times \pi^{-1}(m) : k^{-1}(\{m\} \times \mathbb{R}) \to \{m\} \times \mathbb{R}$$

Because: $E \times_M E = \sqcup_{m \in M} \pi^{-1}(m) \times \pi^{-1}(m)$, $M \times \mathbb{R} = \sqcup_{m \in M} \mathbb{R}$

Another viewpoint is: That $k : E \otimes_M E \to \underline{\mathbb{R}}$, where $\underline{\mathbb{R}}$ is the trivial vector bundle $M \times \mathbb{R} \to M$. This is a bundle morphism.

◇ **Lemma 3.1.3.2** Every vector bundle over a paracompact space admits a (positive-definite) metric.

∘ **Proof** We will prove this in the theory of bundles. □

★ **Corollary 3.1.3.3** Every bundle on smooth manifold admits a bundle metric.

∘ **Proof** By paracompactness of smooth manifold. □

3.2 Tangent Bundles

3.3 Cotangent Bundles

3.4 Frame Bundle

Definition 3.4.0.1 Let M be an n-dimensional manifold. For every $p \in M$, define:

$$Fr_p(TM) := \{(e_1, \dots, e_n) | Span(e_1, \dots, e_n) = T_pM\}$$

with $Fr(M) := \sqcup_{p \in M} Fr_p(M)$; Proejction $\pi : Fr(M) \to M$;

◇ **Lemma 3.4.0.2** As a matter of fact,

$$Fr_p(TM) \cong Iso(\mathbb{R}^n, T_pM)$$

∘ **Proof** ... □

Notice that: $Fr(M)|U \subseteq [\mathfrak{X}(U)]^n$. It contains a Lie-bracket structure, but not over $\mathbb{R}$.
If $span_{\mathbb{R}}(E_1, \dots, E_n)$ is a finite dimensional Lie algebra, then

$$[E_i, E_j] = c_{ij}^k E_k \qquad c_{ij}^k : U \to \mathbb{R}$$

But c_{ij}^k is not constant. If we consider it as $C^\infty(U)$-module, then

$$[E_i, E_j] = c_{ij}^k E_k$$

is actually a 'Lie bracket'.

Einstein's Summation Convention

Sometimes, instead of

$$\sum_i a_i$$

We just use a_i, if there is no confusing.

3.4.1 Principal $GL(n, \mathbb{R})$-Bundle Structure

♠ **Proposition 3.4.1.1** For each fibre Fr_pM, it contains a $GL_n(\mathbb{R})$-transform

$$(e_1, \dots, e_n) \cdot A = (\sum e_j A_{j1}, \dots, \sum_j e_j A_{jn})$$

And such action is: Free and transitive. Thus π is a principal $GL_n(\mathbb{R})$-bundle.

∘ **Proof** We just need to verify, π is locally trivial: Notice that $\pi^{-1}(U) \cong \sqcup_{q\in U} Fr_q(M) \cong U \times GL(n, \mathbb{R})$. The isomorphism is given by: Choose a local coordinate $(U; x^1, \dots, x^n)$, with local frame

$$(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n})$$

Then, we got a section $s : U \to \sqcup_{p\in U} Fr_p(M), p \mapsto (\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n})$, and any frame on p could be represented as

$$(e_1, \dots, e_n) = (\frac{\partial}{\partial x^i})_{i=1}^n \cdot A$$

Thus, $Fr(TM)|U \cong GL_n(\mathbb{R}) \times U$; □

The followings are equivalent:

- Local frame;
- Section $s : U \to \sqcup Fr_p(M)$;
- Local trivialization;

∘ **Proof** As a matter of fact: We could take another viewpoint. Given an original local frame

$$s : U \to Fr(M)|U$$

Then, we got a form of $s(p) = s_0(p)g(p), p \in U$; Thus we obtain a smooth map

$$g : U \to GL_n(\mathbb{R})$$

This obtains a bijection:

$$Fr(M)|U \Leftrightarrow C^\infty(U, GL_n(\mathbb{R}))$$

Now, for the local trivialization: We choose a local frame $s \in Fr(M)|U$, then corresponds $C^\infty(U, GL_n(\mathbb{R}))$.

For selected point $q \in U$, and every $u \in Fr(T_pM)$, there exists decomposition

$$u = s(p)g$$

Thus, $u \mapsto (p, g)$ $\quad Fr(T_pM) \cong (p, GL_n(\mathbb{R})) \Rightarrow Fr(TM) \cong \sqcup_{p\in U}(p, GL_n(\mathbb{R})) = U \times GL_n(\mathbb{R})$ □

So, the isomorphism is not canonically. It depends on choice of local frame.

♠ **Proposition 3.4.1.2** Furthermore, there exists associated bundle structure

$$TM \cong Fr(M) \times_{GL_n(\mathbb{R})} \mathbb{R}^n$$

∘ **Proof**

□

♣ **Example 3.4.1.3** (Frame Bundle of $\mathbb{R}^n$) Especially, the frame bundle of $\mathbb{R}^n$ is trivial:

$$Fr(\mathbb{R}^n) \cong \mathbb{R}^n \times GL_n(\mathbb{R})$$

Because $\mathbb{R}^n$ has a standard basis on each tangent space.

♣ **Example 3.4.1.4** (The Action of $SO(3)$ on TS^2)
For the orthogonal frames, we will discuss the in Riemann geometry;

Geometric Structure

- Riemannian Metrics$\Leftrightarrow O(n)$;
- Oriented Riemann Manifolds$\Leftrightarrow SO(n)$;
- $GL(m;\mathbb{C}) \subseteq GL(2m,\mathbb{R})$: Almost Complex Structure;
- $U(m)$: Almost Hermitian Geometry;
- …

We will research G-structure in other topics.

Transition Function

Definition 3.4.1.5 Now, consider two elements in frame:$p \in U \cap V$, then for $s \in Fr(M)|U, s' \in Fr(M)|V$, we have

$$s|U \cap V = s'|U \cap V \cdot (g_{ij})$$

Such $(g_{ij}) : U \cap V \to GL_n(\mathbb{R})$ is called **transition function**

Coframe

Definition 3.4.1.6 Define the **coframe**: Given $(E_1, \dots, E_n) \in Fr(M)|U$, then the **dual coframe** is the unqiue basis:

$$(\theta^1, \dots, \theta^n) \qquad \theta^i \in \Omega^1(U)$$

such that $\theta^i(E_j) = \delta_{ij}$.

Then, the **coframe** on U is

$$Fr(T^*M)$$

3.4.2 Moving Frame

Definition 3.4.2.1

- A **moving frame** is a local section of frame bundle

$$s : U \to Fr(M)$$

- The **structure constant** of moving frame is (c_{ij}^k) given by:

$$[E_i, E_j] = \sum c_{ij}^k E_k$$

where the Lie bracket is well-defined on $\mathfrak{X}(U)$;

Moving Coframe

♠ **Proposition 3.4.2.2** (Structure Equation of Coframe)Now, consider moving frame $(E_1, \dots, E_n)$, with structure constant

$$[E_i, E_j] = \sum c_{ij}^k E_k$$

,then for the dual coframe $(\theta^1, \dots, \theta^n)$, we have a basis $\{\theta^i \wedge \theta^j\}_{i<j}$ for the local frame $\Lambda^2 T_*M$: Thus for every $d\theta^k \in \Omega^2(U)$,

$$d\theta^k = -\frac{1}{2} c_{ij}^k \theta^i \wedge \theta^j$$

Conversely, we could also obtain the structure constant from .

∘ **Proof** ... □

Moving Frame with Connection

3.5 Orientation Bundle

Definition 3.5.0.1 ...

3.6 Connections without Metric Structures

Definition 3.6.0.1 Let: $\pi : E \to M$ be a smooth vector bundle.

- A **(Koszul) connection** is a linear map

$$\nabla : \Gamma(E) \to \Omega^1(M; E) = \Gamma(T^*M \otimes E) \cong \sqcup_p Hom_{\mathbb{R}}(T_pM, E_p)$$

which satisfies Lebnitz's Law

$$\nabla(fs) = df \otimes s + f\nabla s \qquad f \in C^\infty(M), s \in \Gamma(E)$$

Thus, the value of $\nabla s, s \in \Gamma(E)$could be seen as a morphism between vector bundles

$$\nabla s : TM \to E$$

- Or equivalently, in **covariant derivative**

$$\nabla_{(-)}(-) : \mathfrak{X}(M) \times \Gamma(E) \to \Gamma(E)$$

It is actually a bilinear form, which satisfies:

- $\nabla_{fX+gY} s = f\nabla_X s + g\nabla_Y s$;(linear)
- $\nabla_X(s+t) = \nabla_X(t) + \nabla_X(t)$;(Additive)
- $\nabla_X(fs) = X(f)s + f\nabla_X s$;(Lebnitz)

Notice, it is not bilinear, and

$$\nabla_X(-) : \Gamma(E) \to \Gamma(E) \qquad \nabla_X \in \mathrm{Diff}^1_{\mathbb{R}}(\Gamma(E))$$
$$\nabla_{(-)}(s) : \mathfrak{X}(M) \to \Gamma(E) \qquad s \in \Gamma(E)$$

The two definition corresponds through:

$$(\nabla s)(X) = \nabla_X s \qquad X \in \mathfrak{X}(M) = TM \qquad s \in \Gamma(E) \qquad X \in \mathfrak{X}(M)$$

The followings are equivalent:

- $\Omega^1(M; E)$;
- $\Gamma(T^*M \otimes E)$;
- The morphisms $Mor_{VectBund}(TM, E)$;

And: $\nabla_X(-)$ is actually a differential operator: We will learn this in analysis on manifold, and towards geometric analysis.

♠ **Proposition 3.6.0.2** The map

$$\mathfrak{X}(M) \hookrightarrow Diff^1_{\mathbb{R}}(\Gamma(E)) \qquad X \mapsto \nabla_X$$

is injective.

∘ **Proof** Notice that: For every $f \in C^\infty(M)$, $s \in \Gamma(E)$,

$$0 = (\nabla_X - \nabla_Y)(fs)$$
$$= (X(f)s + f\nabla_X s) - (Y(f)s + f\nabla_Y s)$$

Especially, when $f = 1_M$, $(\nabla_X - \nabla_Y)(1_M) = 0$, thus

$$= (X(f) - Y(f))s$$

s is arbitrary, thus $X(f) = Y(f) \Rightarrow X = Y$. □

Local Frame Calculating

♠ **Proposition 3.6.0.3** Let $\pi : E \to M$ be a vector bundle. If $(e_1, \dots, e_r) \in Fr(M)|U$ is a frame on U, with dual frame $(\theta^1, \dots, \theta^r)$, then for connection $\nabla : \Gamma(E) \to \Omega^1(M; E)$,

$$\nabla e_j = \sum_i \omega^i_j \otimes e_i$$

where $\omega_j^i \in \Omega^1(U) = \Gamma(T^*U)$ 1-form, and

$$\omega_j^i = \sum_k \Gamma_{jk}^i \theta^k$$
$$\Rightarrow \nabla e_j = \sum_i \sum_k \Gamma_{jk}^i \theta^k \otimes e_i$$

∘ **Proof** Omitted. □

Dual Connection

Definition 3.6.0.4 Consider a connection: For bundle $E \to M$,

$$\nabla : \Gamma(E) \to \Omega^1(M; E)$$

Then, the induced connection:

- With pairing

$$< -, - >: E \otimes E^* \to M \times \mathbb{R}$$
$$(e, e') \mapsto (p, < e, e' >) \qquad e \in E_p, e' \in E_p^*$$

We got

$$< -, - >: \Gamma(E \otimes E^*) = \Gamma(E) \otimes \Gamma(E^*) \to C^\infty(M)$$

- Then, define the **dual connection**

$$\nabla : \Gamma(E^*) \to \Omega^1(M; E^*) = \Gamma(T^*M \otimes E^*)$$
$$\alpha \mapsto (s \mapsto X(\alpha(s)) - \alpha(\nabla_X s))$$

It is the unique connection such that

$$X(< \alpha, s >) =< \nabla_X \alpha s > + < \alpha, \nabla_X s >$$

Equivalently,

$$d < \alpha, s >= (\nabla\alpha, s) =< \alpha, \nabla s >$$

♠ **Proposition 3.6.0.5** For the , the followings are equivalent:

- $X(< \alpha, s >) =< \nabla_X \alpha s > + < \alpha, \nabla_X s >$;
- $d < \alpha, s >= (\nabla\alpha, s) =< \alpha, \nabla s >$;

∘ **Proof** ... □

♠ **Proposition 3.6.0.6** The dual connection satisfies the required condition.

∘ **Proof** ... □

Pullback of Connection, Covariant Derivative

Definition 3.6.0.7 Given:

- Vector bundle $\pi : E \to M$;
- Associated connection and covariant derivative

$$\nabla : \Gamma(E) \to \Omega^1(M; E) \qquad \nabla^{(-)}_{(-)} : \mathfrak{X}(M) \times \Gamma(E) \to \Gamma(E)$$

- Smooth map $f : N \to M$

Then, the **pullback** is given by: With induced bundle $f^*\pi : f^*E \to N$ and induced

$$f^*\nabla : \Gamma(f^*E) \to \Omega^1(N; f^*E) \qquad f^*s \mapsto f^*(\nabla s) \qquad s \in \Gamma(E)$$

where $f^* : \Omega^1(M; E) = \Gamma(T^*M \otimes E) \to \Omega^1(N, f^*E) = \Gamma(T^*N \otimes f^*E), \omega \otimes s \mapsto f^*\omega \otimes f^*s$

♠ **Proposition 3.6.0.8** (Universal Property of Pullback) The following diagram commutes:

$$\begin{array}{ccc} \Gamma(E) & \xrightarrow{f^*} & \Gamma(f^*E) \\ \downarrow{\scriptstyle \nabla} & & \downarrow{\scriptstyle f^*\nabla} \\ \Omega^1(M; E) & \xrightarrow{f^*} & \Omega^1(N, f^*E) \end{array}$$

∘ **Proof** Omitted; □

Parallel Transport

Definition 3.6.0.9 Let

- $\pi : E \to M$ be a bundle;
- Connection, covariant derivative ∇;
- Path $\gamma : I = [a, b] \to M$, which is smooth;

Then, we obtain:

- Pullback bundle $\gamma^*\pi : \gamma^*E \to [a, b]$;
- Pullback of connection $\gamma^*\nabla : \Gamma(\gamma^*E) \to \Omega^1(I, \gamma^*E)$;

Notice, $\Gamma(\gamma^*E) \ni V : I \to \gamma^*E$, and TI has a global frame $\frac{\partial}{\partial t}$, T^*I for $\partial t \in T^*I = \Gamma(T^*I \otimes \gamma^*E)$.
Now, define the **covariant derivative**

$$\frac{D^\nabla}{dt} : \Gamma(\gamma^*E) \to \Gamma(\gamma^*E) \qquad V \mapsto (\gamma^*\nabla V)(\partial t)$$

- V is called **parallel along** γ, if $\frac{D^\nabla V}{dt} = 0$.

Namely, we should solve the problem in ODE:

$$\begin{cases} \frac{D^\nabla}{dt} V = 0 \\ V(a) = v \end{cases}$$

♠ **Proposition 3.6.0.10** (Turn into Lienar ODE under Local Frame)

- Under local frame $(e_1, \dots, e_n)$, if

  ...

  Then, the parallel equation become:

  $$\dot{v}^i + \omega^i_j(\dot{\gamma})v^j = 0$$

- And such $v(t)$ is unique.

∘ **Proof**

-
- By ODE uniqueness theorem;

□

Definition 3.6.0.11 Given $v \in E_{\gamma(a)}$, there exists a unique parallel section $V : I \to \gamma^* E$ such that $V(a) = v$.
A **parallel transport** is a linear transformation

$$P^\nabla_\gamma : E_{\gamma(a)} \to E_{\gamma(b)} \qquad P^\nabla_\gamma(v) = v$$

With these preparation, we will discuss holonomy group later.
Remark More general theory, will be given in fibre bundle and Ehresmann connection.

Connection in First Jet Exact Sequences

♠ **Proposition 3.6.0.12** The followings are equivalent:

- A connection $\nabla : \Gamma(E) \to \Omega^1(M; E)$;
- The first jet exact sequence splits

$$0 \to T^*M \otimes E \to J^1E \to E \to 0$$

∘ **Proof** ... □

3.6.1 Principal Bundle from Connection, Pincipal Connection Form

Definition 3.6.1.1 Let $\pi : E \to M$ be a vector bundle with $rank(\pi) = r$, the Koszul connection $\nabla : \Gamma(E) \to \Omega^1(M, E)$
Under frame bundle $P = Fr(E)$

$$P \to M$$

3.6.2 Linear/Affine Connections

Definition 3.6.2.1

- Especially, if $E = TM$, then the connection

$$\nabla : \Gamma(TM) \to \Gamma(T^*M \otimes TM) = \Gamma(End_{\mathbb{R}}(TM)) \cong End_{C^\infty(M)}(\mathfrak{X}(M))$$

is called **linear connection**.

- Under this definition, the covariant derivative: $\Gamma(TM) = \mathfrak{X}(M)$,

$$\nabla_{-}(-) : \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$$

For each $A \in \Gamma(TM \otimes T^*M)$, we have expansion under the frame and dual coframe

$$(e_1, \dots, e_n) \qquad (\theta^1, \dots, \theta^n)$$

Then, $A = \sum A^i_j e_i \otimes \theta^j$;

Furthermore, a moving frame turns a tensor field in a scalar coefficiet functions. The example above is $(1,1)$ tensor field.

★ **Corollary 3.6.2.2** For fixed bundle $\pi : E \to M$, the collection of linear collections $Conn(E)$ is an affine space, and a correspondent vector space could be:

$$\Omega^1(M; End(E))$$

∘ **Proof** ... □

Remark The differential

...

is not a (linear) connection, but actually $C^\infty(M)$-linear:

$$A =$$

Torsion Tensor

Definition 3.6.2.3 Define

- the **torsion**

$$T^\nabla(X,Y) := \nabla_X Y - \nabla_Y X - [X,Y] \qquad \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$$

- the **curvature**

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z \qquad \mathfrak{X}(M) \times \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$$

We will calculate the coordinate form of the two valuables in next secion.

♠ **Proposition 3.6.2.4** (Torsion) For the connection ∇:

- It is anti-symmetric:

...

- Thus, it can be factorized in

$$T \in \Gamma(\Lambda^2 T^*M \otimes TM) = \Omega^2(M; TM)$$

∘ **Proof** ... □

♠ **Proposition 3.6.2.5** (Curvature) For the connection ∇:

- $R(X,Y) = -R(X,Y)$;
- $R \in \Gamma(\Lambda^2 T^*M \otimes T^*M \otimes TM) = \Omega^2(M; End(TM))$;

∘ **Proof** ... □

Definition 3.6.2.6

- If $T(X,Y) = 0$, then **torsion-free**;
- If $R(X,Y) = 0$, then **connection flat**;

Cartan's Structure Theorem I II and Component Expansion

Definition 3.6.2.7 Given a frame bundle $(e_1, \dots, e_n)$ of $TM \to M$ with coframe $(\theta^1, \dots, \theta^n)$, we have

$$\nabla e_j = \omega_j^i \otimes e_i \qquad \omega_j^i \in \Gamma(T^*M) = \Omega^1(U)$$

Then, $\omega = (\omega_j^i) \in \Omega^1(U; \mathfrak{gl}_n(\mathbb{R}))$;

▶ ***Theorem*** (Cartan) For connection ∇ and torsion T^∇, under the frame $(e_1, \dots, e_n)$, and dual frame $(\theta^1, \dots, \theta^n)$, we have

$$T = \sum_i T^i \otimes e_i$$
$$T^i = d\theta^i + \sum_j \omega_j^i \wedge \theta^j$$

∘ **Proof** ... □

▶ ***Theorem*** (Cartan) For connection ∇ and curvature R, under the frame $(e_1, \dots, e_n)$ and $(\theta^1, \dots, \theta^n)$,

$$R(X,Y)e_j = \sum_i (\Omega_j^i(X,Y))e_i \qquad \Omega_j^i = \theta^i(R(-,-)e_j)$$
$$\Omega_j^i = \sum_k d\omega_j^i + \omega_k^i \wedge \omega_j^k$$

∘ **Proof** ... □

3.6.3 Existence of a Connection

Recall:

◇ **Lemma 3.6.3.1** Every paracompact manifold has a unity of partion.

∘ **Proof** In 'general topology' □

▶ ***Theorem*** (Existence of Connection) For smooth bundle $\pi : E \to M$, and M is a paracompact smooth manifold, then:
Every smooth vector bundle admits a connection;

∘ **Proof** ... □

★ **Corollary 3.6.3.2** The collection of connections $Conn(E)$ is an affine space. i.e. for $A' := \nabla' - \nabla$, it is $C^\infty(M)$-linear.

∘ **Proof** ... □

Especially, with condition for Riemann metric g, we have

$$\nabla g = 0 \qquad T^\nabla = 0$$

We got Levi-Civita connection.

3.6.4 Flat Connections, and Twisted De Rham Cohomology

Parallel Frame

Definition 3.6.4.1 Let $\pi : E \to M$ be a vector bundle, with connection ∇:

- A local frame $(e_1, \dots, e_n)$ is called a **parallel frame**, if

$$\nabla e_i = 0$$

♠ **Proposition 3.6.4.2** ...

∘ **Proof** ... □

Flat Connections

Definition 3.6.4.3 Let $E \to M$ be a vector bundle. ∇ is a connection.
∇ is called a **flat connection**, if for any $X, Y \in \mathfrak{X}(M)$

...

Covariant Derivative

Definition 3.6.4.4 Define: The **covariant exterior derivative**

$$d_\nabla : \Omega^k(M; E) \to \Omega^{k+1}(M; E) \qquad \alpha \otimes s \mapsto d\alpha \otimes s + (-1)^k \alpha \wedge \nabla s$$

The following rule

...

is called **graded Lebnitz rule**.
Remarks Sometimes operator d_∇ could be denoted as D, if the context is in Cartan's equation.
♠ **Proposition 3.6.4.5**

- (Derivative)
- (Curvature)

∘ **Proof** ... □

♠ **Proposition 3.6.4.6** (Affine Case of Covariant Exterior Derivative, Torsion)

∘ **Proof** ... □

Twisted De Rham Complex

♠ **Proposition 3.6.4.7**

- When $k = 0$, $d_\nabla s = \nabla s$; Thus, d_∇ is a continuation of connection;
- If ∇ is flat, then

$$d_\nabla^2 = 0$$

thus $(\Omega^k(M; E))$ is a cochain complex.

∘ **Proof** ... □

Twisted De Rham Cohomology

Definition 3.6.4.8

3.6.5 Holonomy Groups

Definition 3.6.5.1 Let $\pi : E \to M$ be a vector bundle with connection ∇. And fix a point $p \in M$:

- Consider the loop space with point $(\Omega M, p)$
  Define:

$$Hol_p(\nabla) := \{P_\gamma | \gamma \in \Omega M, P_\gamma \text{ is Parallel}\} \subseteq GL(E_p)$$

is called the **holonomy group** at point p.

♠ **Proposition 3.6.5.2** It is actually a group, with

$$P_{\gamma_1} \circ P_{\gamma_2} = P_{\gamma_1 \circ \gamma_2}$$

∘ **Proof** ... □

Restricted Holonomy Group

Infinitesimal Holonomy

Flatness

★ **Corollary 3.6.5.3** (Flatness) If $R^{\nabla} = 0$, then

$$Hol_p^0(\nabla) = 1$$

∘ **Proof** ... □

AmbroseSinger theorem

3.6.6 Mappings of Connections

Definition 3.6.6.1 ...

3.6.7 Invariant Connections

3.7 Curvature

In the pasting discussion, we have know: The differential map of chain complex relies on the flatness of connections. But if not, we have

$$d_{\nabla}^2 \neq 0$$

We could do some research of the obstruntion to the connection-induced differential squaring to zero:

Definition 3.7.0.1 Suppose $\pi : E \to M$ be a vector bundle, then,

$$F_{\nabla} := d_{\nabla}^2$$

is called **curvature**.

Affine Connection Case

♠ **Proposition 3.7.0.2** Especially: For the bundle $\tau : TM \to M$, we have

$$R = F_{\nabla}$$

∘ **Proof** ... □

3.7.1 Bianchi Identity I II

Bianchi I

▶ ***Theorem*** (Bianchi) Suppose $E = TM$, ∇ is an affine connection.

∘ **Proof** ... □

Bianchi II

▶ ***Theorem*** (Bianchi)

∘ **Proof** ... □

3.8 Jet Bundles

Remark And in the future, we will work with Chern-Weil theory.

Chapter 4 Submersions, Immersions and Embeddings

4.1 Maps of Constant Rank

4.2 Embeddings

Definition 4.2.0.1 Let M, N be two manifolds. Then, a **smooth embedding** $i : N \to M$ satisfies:

- i is topological embedding;
- i is smooth;

4.2.1 Normal Bundle

Definition 4.2.1.1 Suppose $i : N \hookrightarrow M$ be a smooth embedding, then define the **normal bundle**

$$v_{N/M} := i^*TM/TN$$

Induced by $i^* : TN \to i^*TM$. The bundle structure

$$v_{N/M} \to N$$

Remark The structure of normal bundle here depends on the differential structure(Especially, tangential space), thus we could not generalize this definition to the topological manifold.

An alternative is **microbundle** given by Milnor. We will introduce this in topological manifold.

♠ **Proposition 4.2.1.2** There exists a short exact sequence

$$0 \to TN \xrightarrow{i^*} i^*TM \xrightarrow{p} v_{N/M} \to 0$$

and the sequence splits. i.e. There exists $\iota : v_{N/M} \to i_*TM$ with $p \circ \iota = id_{v_{N/M}}$.

∘ **Proof** We just observe the sequence fibrewise. □

Especially, according to our definition of bundle metric, by choosing a positive-definite metric, we obtain

$$v_{N/M} \cong TN^{\perp}$$

Thus, $i_*TM \cong v_{N/M} \oplus TN$.

Remark This property depends on Ext^1 for the nontrivial extension.(If trivial, then splits, and isomorphic to direct sum)

♠ **Proposition 4.2.1.3** (Embedded in $\mathbb{R}^n$) Suppose $M \subseteq \mathbb{R}^n$, then NM is an embedded submanifold of

$$T\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n$$

∘ **Proof** ... □

4.2.2 Tubular Neighborhood

Instead of $v_{M/N}$, we will use: $v_{M/N}$ as the map and $Tot(v_{M/N}) = Tot(v_i)$ as the total space, N for Base space.

▶ ***Theorem*** Suppose $i : N \hookrightarrow M$ be a smooth embedding. Then, there exists a natural zero-section

$$0_N : N \to Tot(v_{N/M}) = Tot(v_i) \qquad p \mapsto 0_p$$

Then, there exists an open neighborhood $V \subseteq Tot(v_i)$ of 0_N, and smooth embedding $\Phi : V \to M$ such that

$$\Phi \circ 0_N = i$$

i.e. There exists factorization

$$\begin{array}{ccc} N & \xrightarrow{0_N} & \exists V \\ & \searrow & \downarrow \Phi \\ & & M \end{array} \qquad \subseteq Tot(v_{N/M})$$

And $\Phi : V \to W = \Phi(V) \subseteq M$ is a diffeomorphism.

∘ **Proof** ... □

★ **Corollary 4.2.2.1** If $M \subseteq N$, then there exists

---

∘ **Proof** ... □

Now, notice that V is open in $Tot(v_{N/M})$. Then, it could be described as

$$V = \{(x, v) | |v| < \epsilon(x)\} \subseteq Tot(v_{N/M}) = \sqcup_{x \in N} i^* T_{i(x)} M / T_x N$$

Especially, if N is compact, then we obtain that

$$V = \{(x, v) | |v| < \epsilon\}$$

4.3 Submersions

4.4 Smooth Covering Maps

Recall

Definition 4.4.0.1 ...

Chapter 5 Submanifolds

5.1 Embedded Submanifold

♠ **Proposition 5.1.0.1** Suppose $F : M \to N$ smooth, $q \in N$ regular value. Then, for $p \in f^{-1}(q)$,

$$T_p(f^{-1}(q)) = Ker(d_pF)$$

∘ **Proof** ... □

5.1.1 Immersed Submanifold

Definition 5.1.1.1 Immersed submanifold: Image of injective immersion.

♣ **Example 5.1.1.2** $\mathbb{R} \to T, x \mapsto (exp(2\pi ix), exp(2\pi irx))$.
Fact If f is closed injective immersion then $Im(f)$ is an embedded submanifold.

5.1.2 Sard's Theorem

▶ ***Theorem*** (Sard) Suppose:

5.1.3 Reuglar Level Set

Definition 5.1.3.1 Let $f : M \to N$ be a smooth map:

- A point $c \in M$ is called a **regular point** if $df_c : T_pM \to T_pN$ is surjective;
  Elsewise, c is called **critical point**.
- A point $d \in N$ is called a **regular value** if $\Phi^{-1}(c)$ is a collection of regular points;
  Such $\Phi^{-1}(c)$ is called a **regular value set**;

▶ ***Theorem*** (Regular Value Set Theorem)

∘ **Proof** ... □

5.2 Whitney's Embedding Theorem

5.3 Whitney's Approxiamation Theorem

Definition 5.3.0.1 ...

5.4 Transversality

Definition 5.4.0.1

- Let: M be a manifold, and $S, S' \subseteq M$ are embedded submanifolds.
  S, S' are said to **intersect transversely** if
  $$T_pM = T_pS \oplus T_pS'$$
- For a smooth map $F : N \to M$, and $S \subseteq M$ be a submanifold. Then, F is tranverse to F if $T_{F(x)}M = dF_x(T_xN) \oplus T_{F(x)}S$.

♠ **Proposition 5.4.0.2** ...

- **Proof** ... □

★ **Corollary 5.4.0.3** ...

- **Proof** ... □

5.4.1 Graphs

♠ **Proposition 5.4.1.1** (Global Characterization of Graphs)

- **Proof** ... □

♠ **Proposition 5.4.1.2** (Local Characterization of Graphs)

- **Proof** ... □

♠ **Proposition 5.4.1.3** (Parametric Transversality Theorem)

- **Proof** ... □

5.4.2 Transversility Homotopy Theorem

5.5 Properties of Morphisms

Chapter 6 Flows

Chapter 7 Differential Forms

7.1 Differential Form

7.1.1 Linear Algebras

Here is some of review of linear algebra.

Definition 7.1.1.1 Take the following notions: k-multilinear map $T^k(V)$, and symmetric/Alternating $\Lambda^k(V)$. And

$$Alt(T) := \frac{1}{k!}\sum_{\sigma\in S_k} sgn(\sigma)T^\sigma; T \in T^k(V)$$
$$\wedge : \Lambda^k(V)\times\Lambda^l(V)\to\Lambda^{k+l}(V)$$
$$\Lambda(V) := \oplus_{k=0}^n \Lambda_k(V)$$

♠ **Proposition 7.1.1.2** We have

$$\begin{cases} dim\Lambda(V) = 2^d \\ dim\Lambda_k(V) = \binom{d}{k}, 0\le k\le d \end{cases}, dim\wedge_k(V) = \{0\}; k\ge d$$

∘ **Proof** ... □

Univeresal Mapping Property

♠ **Proposition 7.1.1.3** Denote $\varphi : V\times\cdots\times V\to\Lambda_k(V), (v_1,\dots,v_k)\mapsto v_1\wedge\cdots\wedge v_k$. Then, for arbitrary alternating multilinear map h of $V\times\cdots\times V$ into the vector space W. There exists a unique map $\tilde{h} : \wedge_k(V)\to W$ such that

$$\tilde{h}\circ\varphi = h$$

With the following diagram commutes

$$\begin{array}{ccc} \Lambda_k(V) & & \\ {\scriptstyle\varphi}\uparrow & \searrow^{h} & \\ V\times\cdots\times V & \longrightarrow & W \end{array}$$

∘ **Proof** ... □

★ **Corollary 7.1.1.4** Suppose that $W = F$. Then, we could take the notion of

$$\wedge_k(V)^* \cong A_k(V)$$

∘ **Proof** ... □

Pairing

Definition 7.1.1.5 Let V and W be real finite dimensional vector spaces, a **pairing** of V and W is a bilinear map

$$(-,-) : V \times W \to \mathbb{R}$$

A **pairing** is called **non-sigular** if whenever $w \neq 0$ in W, theere exists

Linear Transformation of $\Lambda(V)$

Take the notion

$$End(\Lambda(V))$$

for the endomorphisms of $\Lambda(V)$

7.1.2 Tensor Fields and Differential Forms

Differential Forms

Definition 7.1.2.1 Define:

$$\Lambda^k M := \sqcup_{p\in M} \Lambda^k(T_p M);$$

The k-forms locally written as:

$$\omega = \sum_I \omega^I dx^I \qquad dx^I = dx^{i_1} \wedge \cdots \wedge dx^{i_k} \qquad 1 \leq i_1 < \cdots < i_k \leq n$$

Take the following notion $\Omega^k(M)$

And action

$$dx^{i_1} \wedge dx^{i_k}(\frac{\partial}{\partial x_{j_1}}, \ldots, \frac{\partial}{\partial x^{i_k}}) = \delta^I_J$$

7.1.3 Exterior Derivative

Definition 7.1.3.1 Define: For k-form $\omega = \sum_J \omega_J dx^J dx^J$, then

$$\begin{aligned} d\omega &:= \sum d\omega_J \wedge dx^J \\ &= \sum_J \sum_i \frac{\partial_J}{\partial x^i} dx^i \wedge dx^J \end{aligned}$$

♠ **Proposition 7.1.3.2**

- d is linear over $\mathbb{R}$;
- Given k-form ω, l-form η, $U \subseteq \mathbb{R}^n$ or $\mathbb{H}^n$
- $d^2 = 0$;
- d commutes with pullback

$$F^*(d\omega) = d(F^*\omega)$$

∘ **Proof** ... □

▶ ***Theorem*** (Existence and Uniqueness of Exterior Differentation)

∘ **Proof** ... □

Closed,Exact Form

Definition 7.1.3.3 Consider the cohomology: $\cdots \to H^{k+1}_{dR}(M) \to H^{k}_{dR}(M) \to \dots$. Then, the differential form $\omega \in \Omega(M)$ is:

- Closed, if $d\omega = 0$;
- Exact, if there exsits some f such that $\omega = df$;

Especially, for 1-form:

♠ **Proposition 7.1.3.4** For 1-form $\omega = \sum a_i dx^i$, we have

$$d\omega = \sum (\frac{\partial a_i}{\partial x_i} - \frac{\partial a_j}{x_j}) dx^i \wedge dx^j$$

Thus closedness$\Leftrightarrow \frac{\partial a_i}{\partial x_i} = \frac{\partial a_j}{\partial x_j}$

7.1.4 The Lie Derivative

7.2 Orientation

7.2.1 Orientation of Vector Spaces

Definition 7.2.1.1 Let V be an N-dimensional real inner product space. We extend the inner product from V to all of $\Lambda(V)$

$$< w_1 \wedge \cdots \wedge w_p, v_1 \wedge \cdots \wedge v_p > := det < w_i, v_j >$$

The bilinear map can be extended linearly to $\Lambda_p(V)$.

♠ **Proposition 7.2.1.2** If $e_1, \dots, e_n$ is an orthonormal basis of V, then the corresponding basis of $\Lambda(V)$ is an orthonormal basis for $\Lambda(V)$.

∘ **Proof** ... □

Definition 7.2.1.3 (Consistently Oriented) Assume that $(v_1, \dots, v_n)$ and $(w_1, \dots, w_n)$ be two bases of V with $dim(V) = n$, then we have the transform matrix

$$v^T = Bw^T$$

Then, they are **consistently oriented** if $det(B) > 0$.

7.2.2 Orientation of Manifold

Definition 7.2.2.1 Let M be a manifold. Then a frame

$$(E_1, \dots, E_n)$$

is **(positively) oriented** if

7.2.3 The Orientation Covering

7.2.4 The Orientation of Hypersurface

7.2.5 The Orientation of

Chapter 8 Integration on Manifolds

8.1 Integration on Manifolds

8.1.1 Euclidean Space

Definition 8.1.1.1 Defien:

$$f : D \to \mathbb{R} \qquad \int_D f\,dx^1 \wedge \cdots \wedge dx^n := \int_D f\,dx^1 \ldots dx^n$$

8.1.2 Manifolds

Definition 8.1.2.1 Let M be a manifold. A **partition of unity** is a collection of smooth maps $(\varphi_i : M \to \mathbb{R})$

- there exists $\varphi_i : U_i \to V_i$ such that

$$supp(\varphi_i) \subseteq U_i$$

- For each $p \in M$, only finitely many of $\varphi_i(p) \neq 0$.

Definition 8.1.2.2 Let M be an oriented n-dim smooth manifold, ω a diff n-form, local integrable. Choose a partition of unity (φ_i), define

$$\int \omega = \sum_i \int h_i^{-1*}(\varphi_i \circ \omega)$$

Chapter 9 De Rham Cohomology

9.1 De Rham Complex

Chapter 10 Integratability

10.1 Integral Curve

Definition 10.1.0.1 Let: M be a smooth manifold, $X \in \mathfrak{X}(M)$.
A **integral curve** is a smooth curve $\gamma : I \to M$ such that, for every $t \in I$,

$$\dot{\gamma}(t) = X_{\gamma(t)}$$

where $\dot{\gamma}(t) = (d\gamma)_t(\frac{\partial}{\partial t}) \in T_{\gamma(t)}M$

Sometimnes we will use the notion $d_{\gamma_s}(1)$, because $1 \in \mathbb{R}$ corresponds the standard tangent vector $\frac{\partial}{\partial t}|s$.
Or, another explaination is to consider the curve goes through t: $c(r) = t + r$, $d\gamma_s(1) = d\gamma_s(c'(0)) = (\gamma \circ c)'(0) = \frac{d}{dr}|_{r=0}\gamma(s + r)$.

10.1.1 Existence of Maximal Integral Curve

♠ **Proposition 10.1.1.1** (Frame Coordinate of Integral Curve) Take frame bundle $(x^1, \dots, x^n)$, and $X \in \mathfrak{X}(M)$, then the equation of integral curve at p with coordinate $\varphi = (x^1, \dots, x^n) : U \to \mathbb{R}^n$, then

$$X|U = \sum X^i(x)\frac{\partial}{\partial x^i}$$
$$x(t) = \varphi(\gamma(t))$$

We obtain

$$\dot{x(t)} = (X^1(x(t)), \dots, X^n(x(t)))$$

Especially, it is local-Lipschitz.

∘ **Proof** ... □

♠ **Proposition 10.1.1.2** (Local Existence) For each point $p \in M$, there exists a local integral curve γ which goes through p.

∘ **Proof** Apply Picard-Lindelof. □

10.1.2 Maximal Integral Curve

Definition 10.1.2.1 Let $X \in \mathfrak{X}(M)$. Now, consider the inclusion of curves:

$$(\gamma : I \to M, I) \subseteq (\theta : J \to M, J) \Leftrightarrow i \subseteq J, \theta|I = \gamma$$

♠ **Proposition 10.1.2.2** There exists a unique maximal curve which goes through p. (We could take $\gamma_p(0) = p$)

∘ **Proof** By uniqueness of ODE. □

10.1.3 Visualization of Integral Curve

10.1.4 Compelete Vector Field

Definition 10.1.4.1 A vector field $X \in \mathfrak{X}(M)$ is called **compelete**, if for each $p \in M$, there exists a unique maximal integral curve

$$\gamma_p : I_p \to M, \gamma_p(0) = p \qquad I_p = \mathbb{R}, \forall p \in M$$

10.2 Distributions and Involutivity

10.3 Frobenius Theorems

Chapter 11 Appendix: Smooth Category

Definition 11.0.0.1 ...